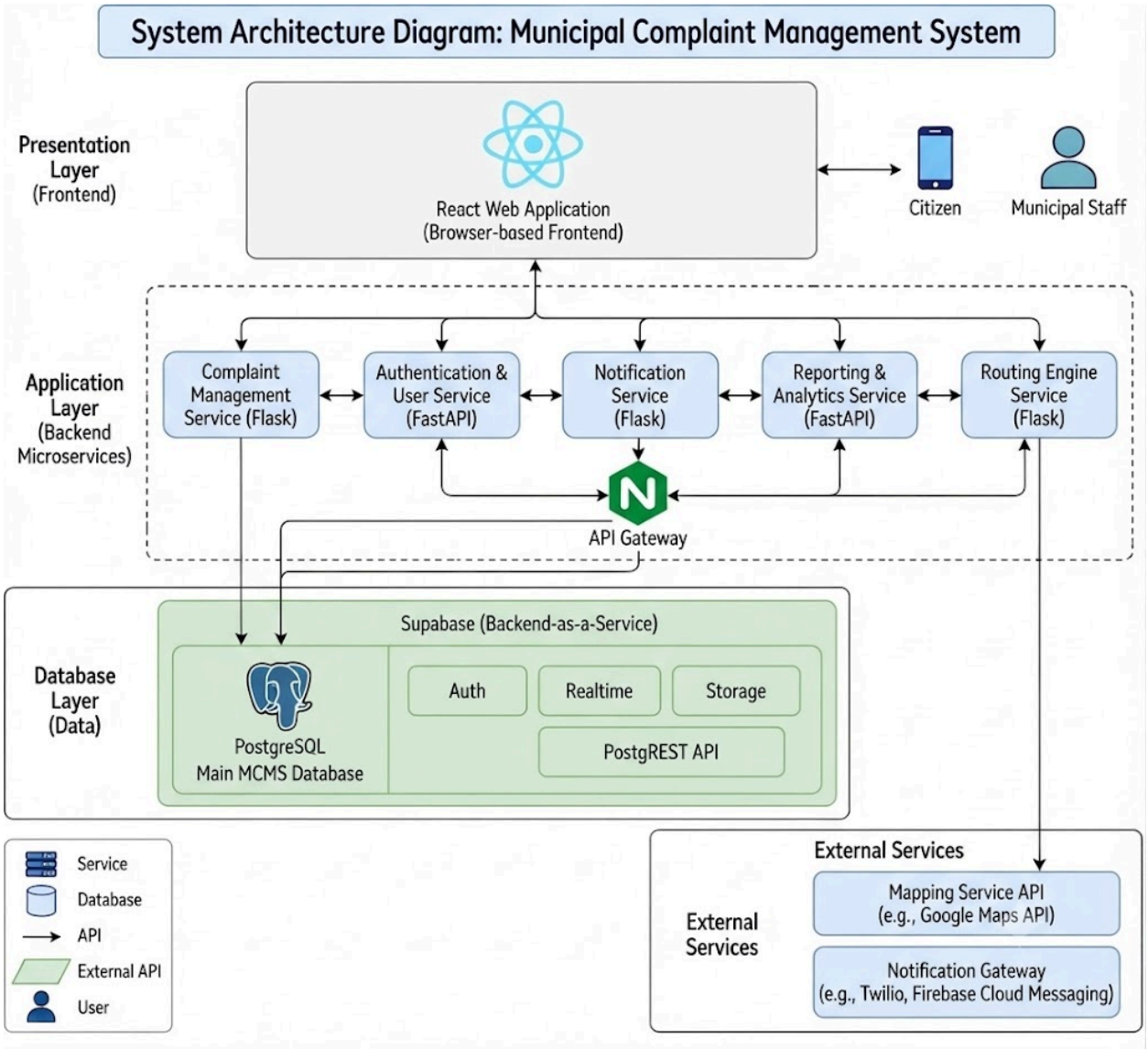


1. System Architecture Overview

The Municipal Complaint Management System (MCMS) follows a robust, multi-tier distributed architecture pattern designed to ensure high availability, clean separation of concerns, scalability, and secure data handling. The architecture is broadly divided into four primary operational layers:

- **Presentation Layer (Frontend):** Manages user interactions for citizens and municipal staff.
- **Application Layer (Backend Microservices):** Handles core business logic, routing, and processing via microservices.
- **Database Layer (Data):** Manages relational data persistence, authentication, and secure access policies.
- **External Services:** Integrates third-party APIs for mapping, geolocation, and notifications.

2. Detailed Component Breakdown & Theory

A. Presentation Layer (Frontend)

- **Technology Stack:** React Web Application / Progressive Web App (PWA).
- **Core Responsibility:** Acts as the primary user interface (UI) and entry point for users. It allows citizens to lodge complaints with geo-tagged images and track their status in real-time, while providing municipal officials and field workers dedicated dashboards to review, update, and resolve tickets securely via HTTPS REST/JSON APIs.

B. Application Layer (Backend Microservices)

- **Technology Stack:** Python (Flask & FastAPI) managed behind an API Gateway (Nginx).
- **Core Microservices:**
    1. **Complaint Management Service (Flask):** Handles grievance creation, state transitions, and tracking lifecycle updates.
    2. **Authentication & User Service (FastAPI):** Manages user registration, secure session tokens, and Role-Based Access Control (RBAC) for Citizens, Municipal Officers, and Field Staff.
    3. **Routing Engine Service (Flask):** Automatically categorizes and routes incoming complaints to the appropriate municipal department based on location and grievance type.
    4. **Notification Service (Flask):** Triggers real-time alerts and status updates to users as tickets progress through different stages.
    5. **Reporting & Analytics Service (FastAPI):** Compiles performance metrics, SLA escalations, and resolution statistics for municipal administrators.

C. Database Layer (Data Management)

- **Technology Stack:** Supabase (Backend-as-a-Service powered by PostgreSQL).
- **Core Components:**
    - **Main MCMS Database:** Stores structured relational tables for users, complaints, departments, and audit logs.

    - **Supabase Auth & Realtime:** Provides secure credential management and instant WebSocket-based UI updates whenever a complaint's status changes.
    - **PostgREST API & Storage Buckets:** Manages secure, cloud-hosted storage for uploaded complaint photos and proof-of-resolution documents.

D. External Services Integration

- **Mapping Service API (e.g., Google Maps API):** Integrated to capture precise geolocation coordinates and display complaint hotspots on maps.
- **Notification Gateway (e.g., Twilio / Firebase Cloud Messaging):** Dispatches automated SMS and push notifications to citizens and field workers regarding ticket updates and closures.